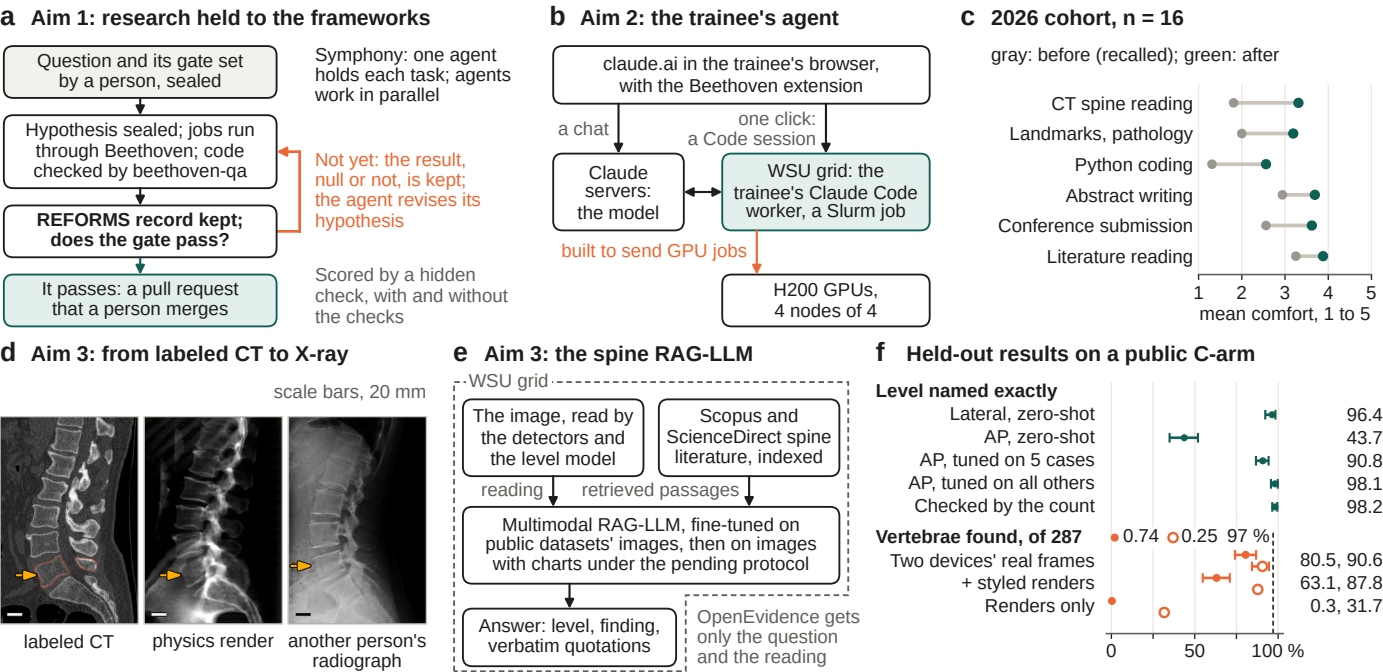

a Aim 1: research held to the frameworks
Question and its gate set by a person, sealed
Hypothesis sealed; jobs run through Beethoven; code checked by beethoven-qa
REFORMS record kept; does the gate pass?
It passes: a pull request that a person merges
Symphony: one agent holds each task; agents work in parallel
Not yet: the result, null or not, is kept; the agent revises its hypothesis
Scored by a hidden check, with and without the checks
b Aim 2: the trainee's agent
claude.ai in the trainee's browser, with the Beethoven extension
a chat
one click: a Code session
Claude servers: the model
WSU grid: the trainee's Claude Code worker, a Slurm job
built to send GPU jobs
H200 GPUs, 4 nodes of 4
c 2026 cohort, n = 16
gray: before (recalled); green: after
CT spine reading
Landmarks, pathology
Python coding
Abstract writing
Conference submission
Literature reading
1
2
3
4
5
mean comfort, 1 to 5
d Aim 3: from labeled CT to X-ray
scale bars, 20 mm
labeled CT
physics render
another person's radiograph
e Aim 3: the spine RAG-LLM
WSU grid
The image, read by the detectors and the level model
Scopus and ScienceDirect spine literature, indexed
reading
retrieved passages
Multimodal RAG-LLM, fine-tuned on public datasets' images, then on images with charts under the pending protocol
Answer: level, finding, verbatim quotations
OpenEvidence gets only the question and the reading
f Held-out results on a public C-arm
Level named exactly
Lateral, zero-shot 96.4
AP, zero-shot 43.7
AP, tuned on 5 cases 90.8
AP, tuned on all others 98.1
Checked by the count 98.2
Vertebrae found, of 287
0.74
0.25
97 %
Two devices' real frames 80.5, 90.6
+ styled renders 63.1, 87.8
Renders only 0.3, 31.7
0
50
100 %